

Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using data from 3,900 customers across four product categories (Clothing, Accessories, Footwear and Outerwear). The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

The analysis is framed by one overarching business question: *“How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?”* To answer it, the workflow moves from data preparation in Python, to structured analysis in SQL (MySQL), to an interactive Power BI dashboard, ending with business recommendations.

2. Dataset Summary

- Rows: 3,900
- Columns: 18
- Key Features:
 - Customer demographics (Age, Gender, Location, Subscription Status)
 - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
 - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type, Payment Method)
- Missing Data: 37 values in Review Rating column
- Final table after cleaning: 19 columns (2 engineered features added, 1 redundant column removed)

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using `pandas`.
- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.

Column	count	mean	std	min	25%	50%	75%	max
Age	3900	44.07	15.21	18	31	44	57	70
Purchase Amount (USD)	3900	59.76	23.69	20	39	60	81	100
Review Rating*	3863	3.75	0.72	2.5	3.1	3.8	4.4	5.0
Previous Purchases	3900	25.35	14.45	1	13	25	38	50

*Numeric columns – df.describe(). *Before imputation (37 missing values).*

Column	unique	top	freq
Gender	2	Male	2652
Item Purchased	25	Blouse	171
Category	4	Clothing	1737
Location	50	Montana	96
Size	4	M	1755
Color	25	Olive	177
Season	4	Spring	999
Subscription Status	2	No	2847
Shipping Type	6	Free Shipping	675
Discount Applied	2	No	2223
Promo Code Used	2	No	2223
Payment Method	6	PayPal	677
Frequency of Purchases	7	Every 3 Months	584

Categorical columns – df.describe(include='all').

- **Missing Data Handling:** Checked for null values and imputed the 37 missing values in the **Review Rating** column using the median rating of each product category.
- **Column Standardization:** Renamed columns to **snake case** for better readability and documentation (for example, **Purchase Amount (USD)** became **purchase_amount**).
- **Feature Engineering:**
 - Created **age_group** column by binning customer ages into four quartile-based groups: Young Adult (18–31), Adult (32–44), Middle-aged (45–57) and Senior (58–70).
 - Created **purchase_frequency_days** column by converting **frequency_of_purchases** into days (Weekly = 7, Fortnightly / Bi-Weekly = 14, Monthly = 30, Quarterly / Every 3 Months = 90, Annually = 365).
- **Data Consistency Check:** Verified that **discount_applied** and **promo_code_used** were identical in every row, and dropped **promo_code_used** as redundant.
- **Database Integration:** Connected the Python script to MySQL using **SQLAlchemy** and **PyMySQL** and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in MySQL to answer key business questions:

1. Revenue by Gender – Compared total revenue generated by male vs. female customers.

	gender	revenue
1	Female	75191
2	Male	157890

2. High-Spending Discount Users – Identified customers who used discounts but still spent at or above the average purchase amount.

	customer_id	purchase_amount
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62

Total rows: 839 (first 10 shown). Average purchase amount used as the threshold: \$59.76.

3. Top 5 Products by Rating – Found products with the highest average review ratings.

	product	average
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

T-shirt and Handbag also round to 3.78, so the fifth position is a three-way tie at two decimals; Skirt is highest before rounding.

4. Discount-Dependent Products – Identified the 5 products with the highest percentage of discounted purchases.

	item_purchased	discount_rate
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

5. Customer Segmentation – Classified customers into New (1 previous purchase), Returning (2–10) and Loyal (more than 10) segments based on purchase history.

	customer_segment	number_of_customers
1	Loyal	3116
2	New	83
3	Returning	701

6. Top 3 Products per Category – Listed the most purchased products within each category using a ROW_NUMBER() window function.

	item_rank	category	item_purchased	total_orders
1	1	Accessories	Jewelry	171
2	2	Accessories	Belt	161
3	3	Accessories	Sunglasses	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

Outerwear contains only two products. Items with equal order counts (e.g., Belt and Sunglasses) may appear in either order.

7. Repeat Buyers & Subscriptions – Counted customers with more than 5 previous purchases by subscription status to check whether repeat buyers are more likely to subscribe.

	subscription_status	repeat_customers
1	No	2518
2	Yes	958

8. Revenue by Age Group – Calculated the total revenue contribution of each age group.

	age_group	total_revenue
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

5. Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present insights visually. It combines KPI cards with charts for subscription share, category and age-group performance, and slicers for Subscription Status, Gender, Category and Shipping Type.



Dashboard view with no slicer filters applied (all 3,900 customers).

6. Business Recommendations

- **Boost Subscriptions** – Only 27% of customers (1,053 of 3,900) are subscribed. Promote exclusive benefits for subscribers to grow this group.
- **Customer Loyalty Programs** – 701 customers (18%) are still in the “Returning” segment and 83 (2%) are new, versus 3,116 (80%) who are loyal. Reward repeat buyers to move them into the “Loyal” segment; repeat buyers already make up 91% of subscribers (958 of 1,053) versus 88% of non-subscribers (2,518 of 2,847).
- **Review Discount Policy** – 43% of purchases (1,677) used a discount, and Hat, Sneakers and Coat are discounted on roughly half of their sales. Since 839 discount users still spent at or above the average, balance sales boosts with margin control.
- **Product Positioning** – Highlight top-rated products (Gloves, Sandals, Boots, Hat) and best-sellers (Jewelry, Blouse, Pants) in campaigns.
- **Targeted Marketing** – Focus efforts on the highest-revenue age groups: Young Adults lead with \$62,143 (27% of \$233,081 total revenue), followed by the Middle-aged group with \$59,197.